

CITA 216 – Blender Portfolio

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Portfolio Overview

- Highlight key labs and hands-on learning assignments
- Provide a goal overview for each lab/hands-on assignment
- Provide learning outcomes for each lab/hands-on assignment
- Provide a reflection on the following:
 - Project I'm most proud of
 - Biggest technical challenge
 - Tools & features that surprised me
 - Room for improvement

Lab 1: First Steps

- **Goal:** Explore the fundamentals of Blender
- **Learning Outcomes:**
 - Adding 3D primitives into a Blender scene
 - Using basic Blender tools (move, rotate, scale) to position objects in a coherent manner
 - Apply basic materials to objects
 - Utilize Blender's light objects for lighting simulations



Lab 2: Lighthouse on a Rocky Island

- **Goal:**
 - Design a low-poly island with a lighthouse resting on its shore
- **Learning Outcomes:**
 - Use Blender's tools (extrude, insert face, & loop cuts) to develop simple structures from primitives
 - Use sculpting tools to mold the primitive shape into an island. Used tools like dynamic topology to generate vertices, edges, and faces while sculpting
 - Apply the decimate modifier to purposely reduce the smoothness of the island into a low-poly design



Lab 3: Designing a Chess Board

- Goal:
 - Design a chessboard with pawns and at least three unique chess pieces (King, Queen, Rook, Bishop, and Knight)
- Learning Outcomes:
 - Greater understanding of manipulating primitive shapes to generate realistic objects. This includes using smoothing and subdivision surface modifiers to smooth out objects
 - Greater depth in generating shaders to manipulate objects' appearances.
 - Using material slots to apply materials to different faces in the same object



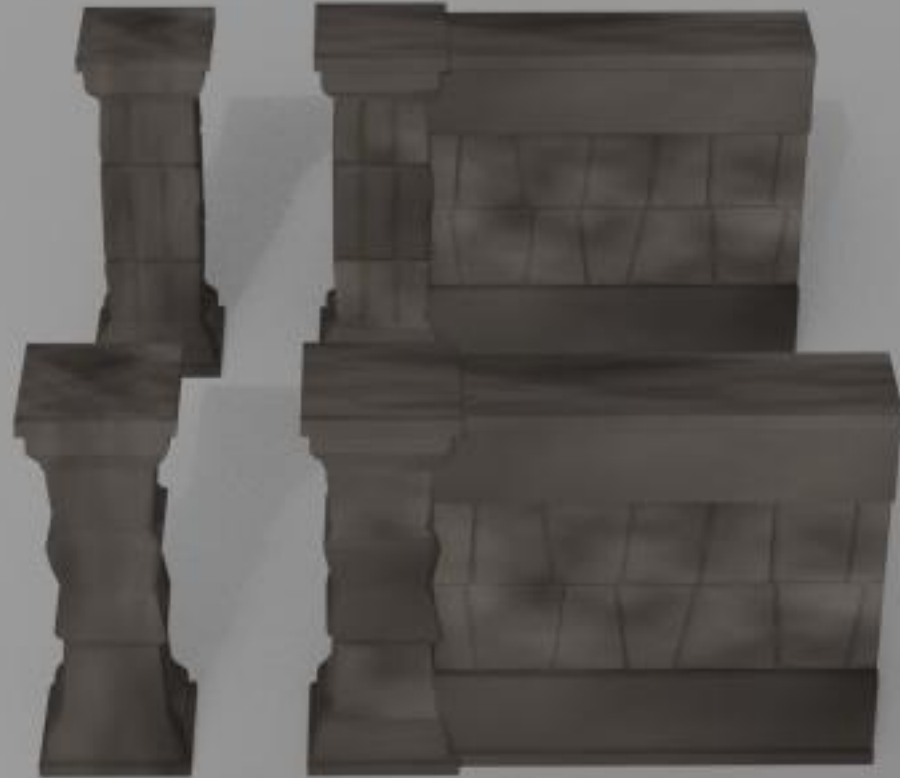
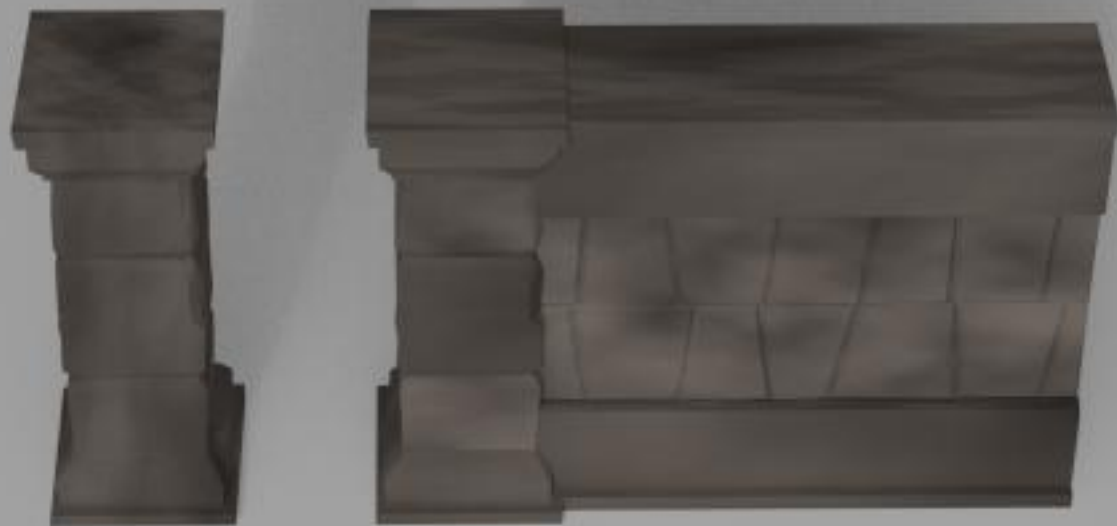






Lab 4 & 5: Low-Poly Dungeon Modeling

- **Goals:**
 - Develop objects to insert into a dungeon layout
 - Create wall and pillar components to create the walls of the dungeon
 - Create a dungeon layout with objects, lighting, walls, and floors
- **Learning Outcomes:**
 - Learning the mindset of a component-based world design
 - Greater proficiency with Blender's modeling tools (knife cutting, bevel vertices & edges)
 - Generating room structures from wall components











Hands On 8: Rigging a Hand

- Goal:
 - Rig a hand with bones and joints to create hand gestures
- Learning outcomes:
 - Use an image references to generate a believable hand
 - Add an armature rig to the hand mesh object
 - Create a hand bone structure
 - Rotate the bones to create hand gestures



Final Group Project

- The final group project consisted of taking concepts learned in labs and hands-on assignments and applying them in one final project
- Final Project Goals
 - Modular Scene Design
 - Rigged Character
 - Interactions & Animations
 - Textures, Lighting, & Composition

Reflection: My Best Achievements

- Lab 4 & 5: low-poly dungeon model
 - First lab where Blender tools and techniques became intuitive
 - Practical application for video game development

Reflection: Biggest Technical Challenge

Animations and rigging

- Rigging objects performed unintended deformations in certain poses
- Lots of steps to get the desired result from rigging
- Using tools and technology for animations was cumbersome

Reflection: Tools & Features that Surprised me

Modifiers

- Modifiers are extremely powerful and allow great control over an object's appearance
- Modifiers work like magic. How can it compute so many scenarios correctly?

Reflection: Room for Improvement

Animations and Rigging

- Compared to other concepts, covering animations and rigging felt more like a crash course
- I was unsuccessful in getting animations to work in lab assignments. I was more successful in the final group project, but there's plenty of room for improvement.
- My plan moving forward is to learn more about Blender from the GameDev.tv Blender bundle. They have a full course on animations and rigging alone.